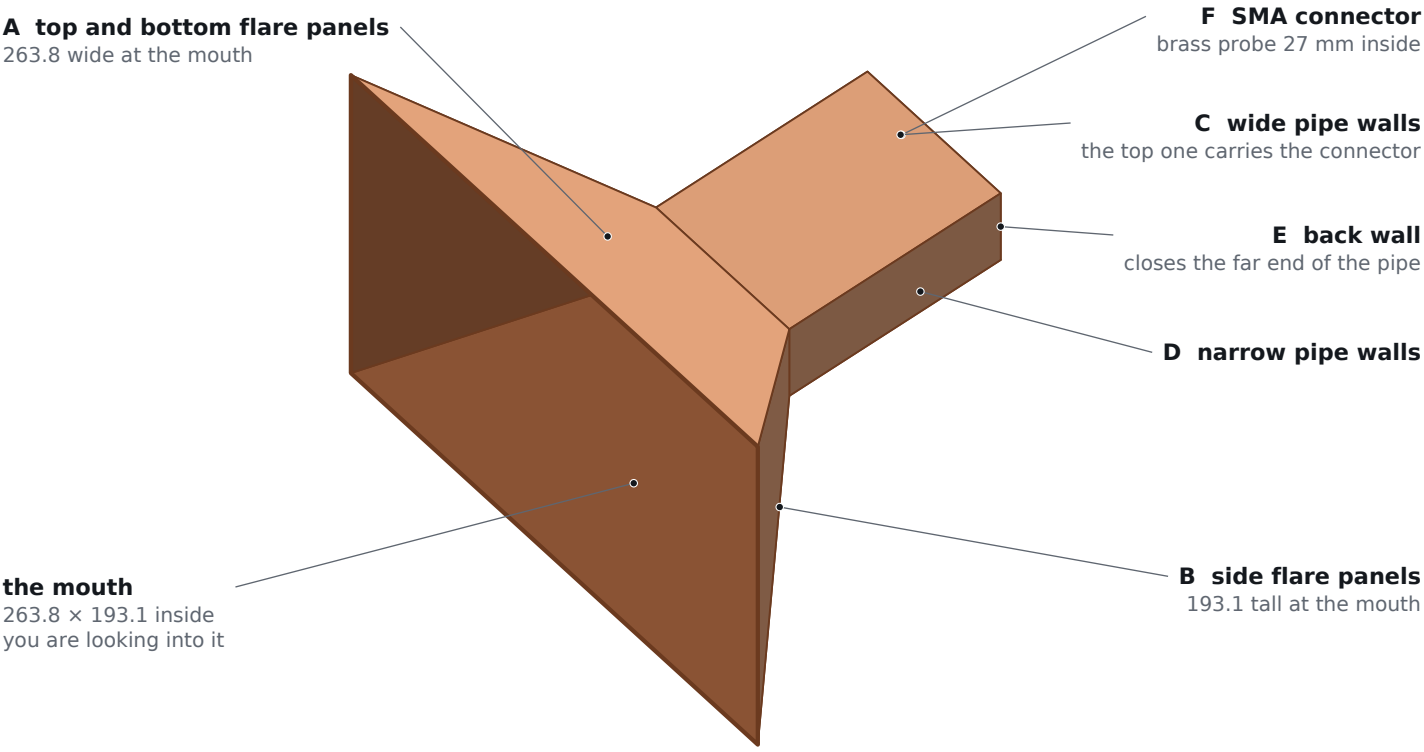
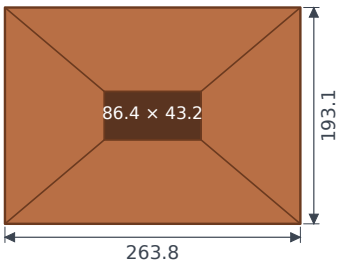
Horn antenna build guide

Three identical horns, cut from 24 gauge copper sheet and soldered together. One transmits, one receives, and one is the spare for the azimuth upgrade.



mouth, inside	263.8 × 193.1 mm
flare depth	91.5 mm, mouth to pipe
pipe, inside	86.4 × 43.2 mm, 115 mm long
probe	27 mm into the pipe, 17.7 mm from the back wall, on the centre line
what it does	focuses the signal about 22 times (13.4 dBi) into a beam about 34° × 36° wide
copper	one 12 × 24 in sheet per horn

looking into the mouth

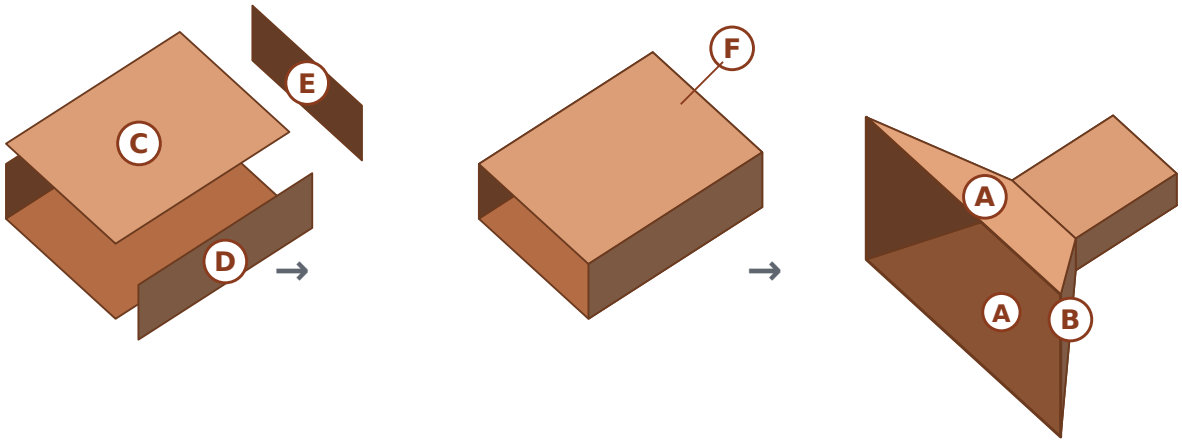


How to use this guide

- Pages 2 and 3 show how the nine pieces fit together, and what the inside looks like.
- Pages 4 to 6 are the shopping list, the cut list, every piece drawn flat, and how to lay one horn out on one sheet.
- Pages 7 to 9 are the build, nine steps, each with its check.
- The last page is a record to fill in for all three horns.

How it fits together

The horn is a short rectangular pipe that opens out into a wide mouth, like a megaphone. The pipe is closed at the back. The radar's cable plugs into a connector on top of the pipe, and a short brass rod under that connector passes the signal into the pipe. The flare then shapes it into a beam. Nine flat pieces make one horn, lettered A to E all through this guide, and F is the bought connector.



1 the pipe
two wide walls C, two narrow walls D, and the back wall E

2 the connector
F bolts to the top wall, its rod reaching down inside

3 the flare
two A panels and two B panels on the front of the pipe

The order it goes together. The pipe is built first as a closed box, then the connector goes on, then the four flare panels.

Words used in this guide

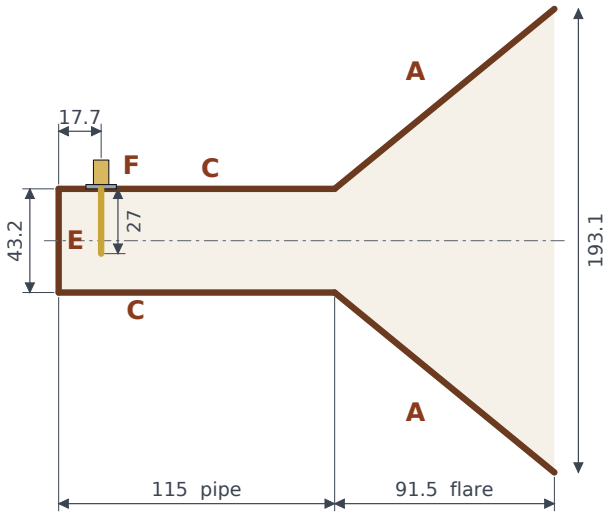
word	means
pipe	the closed rectangular box at the back, 86.4 × 43.2 inside. Pieces C, D and E
flare	the four sloping panels that open out from the pipe to the mouth. Pieces A and B
mouth	the big open end the beam comes out of
probe	the short brass rod inside the pipe, soldered to the connector. It passes the signal between the cable and the horn
tab	a 6 mm strip along an edge, folded over so two pieces overlap and can be soldered
collar	the front tabs of the pipe, folded outward. The flare panels sit on them
seam	any joint between two pieces. Every seam must be soldered closed
inside size	measured between the inside faces of the copper. Every pipe and mouth size in this guide is an inside size

Why the inside matters and the outside does not

The radio wave only ever touches the inside surface of the copper. Keep the inside smooth, every seam closed and every size right. All soldering is done from the outside, and it can look as rough as it likes.

Inside the horn, to scale

Both views cut the horn in half down the middle. Every inside size the build has to hit is on one of them.

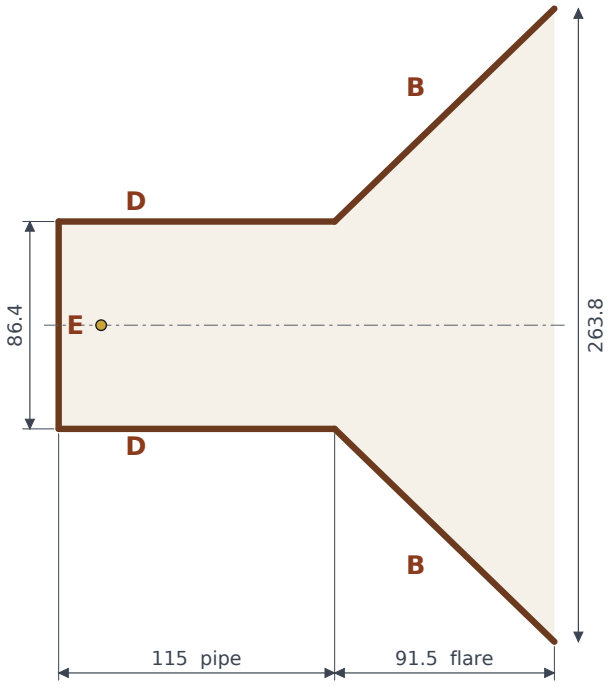


SIDE VIEW, cut down the middle

The connector (F) sits on top of the pipe. Its brass rod hangs 27 mm down into the pipe, 17.7 mm in front of the back wall (E).

The top and bottom panels (A) lean out 39° from the pipe.

Side view. The connector's rod stands straight down from the top wall.



TOP VIEW, cut across the middle

The rod sits exactly halfway across the pipe, on the centre line (the brass dot).

The side panels (B) lean out 44° from the pipe.

Top view. The rod is halfway across, on the centre line.

What you need

for three horns		notes
copper sheet, 24 gauge (0.021 in), 12 × 24 in	3 sheets	one sheet per horn (page 6). Copper, so the seams take solder
SMA female connector, 4-hole flange, solder cup	3, plus spares	must be solder cup, not a PCB pin
brass rod, 1/16 in	about 100 mm	three probes cut 32 mm long, trimmed later
plumbing solder and paste flux		not electronics rosin-core. The seams are long and copper soaks up heat
screws and nuts, M2.5 or 2-56	12 sets	four per connector. The flange holes are too small for M3

tools	used for
aviation snips or a sheet nibbler	cutting the pieces
steel rule, square, fine marker or scriber	marking out
60 to 100 W iron with a big tip, or a small torch	soldering. A 25 W electronics iron will not heat the sheet
two lengths of angle iron or hardwood, and clamps	folding tabs on a straight line
drill, 2.7 mm and 4.5 mm bits, a centre punch, a file	the connector holes, and deburring
calipers and a multimeter	checking sizes and seams
gloves and safety glasses	cut copper edges are sharp. Ventilate while soldering

Cut list, per horn

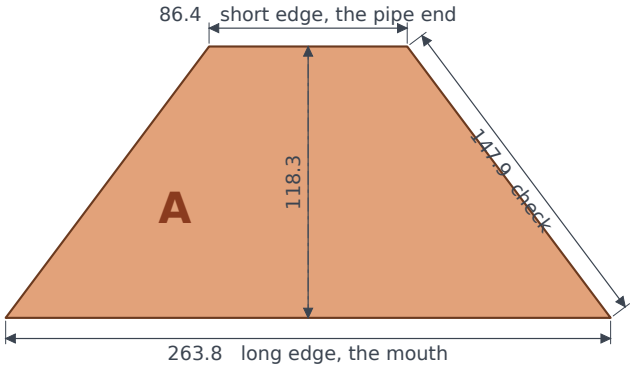
piece	name	qty	size of the wall	tabs	folds
A	flare panel, top and bottom	2	trapezoid, edges 86.4 and 263.8, 118.3 high	none	none
B	flare panel, sides	2	trapezoid, edges 43.2 and 193.1, 127.5 high	both slanted edges	64°
C	pipe wall, wide	2	87.5 × 115	both long edges, and the front edge	90° sides, 39° front
D	pipe wall, narrow	2	43.2 × 115	front edge	44° front
E	back wall	1	88.5 × 44.3	all four edges, corners notched	90°
F	connector and probe	1	bought		

Every tab is 6 mm wide. The sheet is 0.53 mm thick, which is why C and E are cut a little over the pipe's 86.4 mm: the wide walls lap over the edges of the narrow ones, and the back wall fits over all four. The inside still comes out 86.4 × 43.2.

The pieces, flat

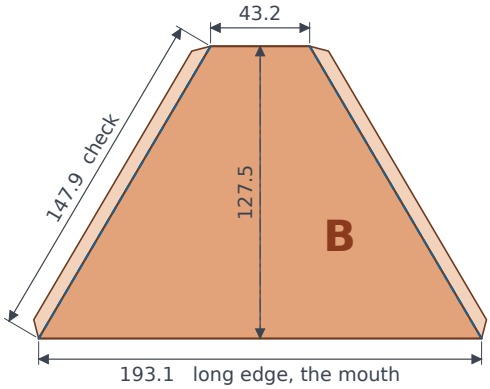
Drawn to scale, each as seen from its outside face. Mark and cut exactly these outlines. Dashed blue lines are where tabs fold.

the wall itself · solder tab, 6 mm · fold line · cut line · sizes in mm, drawn to scale



A top and bottom flare panels
cut 2 per horn · no tabs

The long edge is the mouth, the short edge meets the pipe. The height is measured up the centre line, square to both edges.

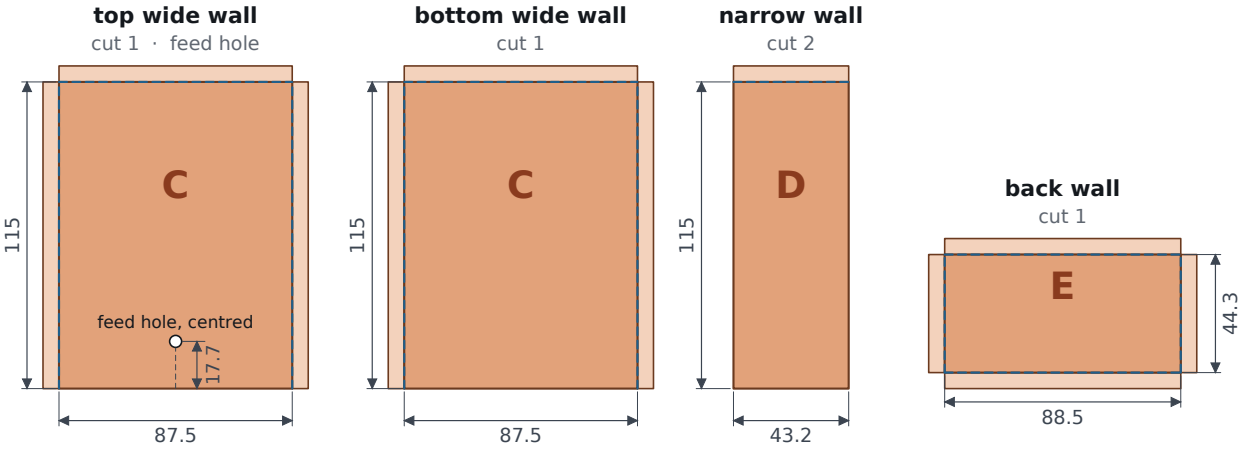


B side flare panels
cut 2 per horn

A 6 mm tab on both slanted edges, its ends cut off at 45°.

Fold each tab 64° away from you, so it lies on the outside of an A panel.

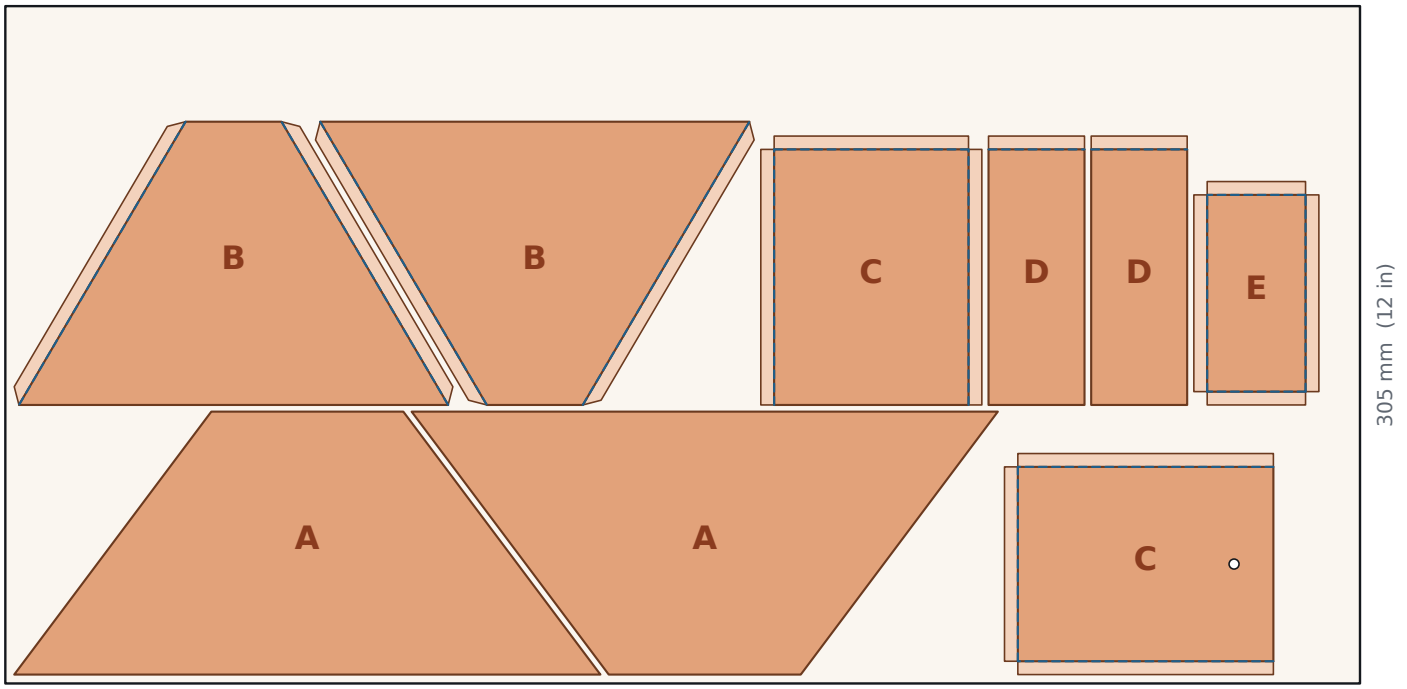
The slanted edges of A and B must both come out 147.9 mm, because that is where they meet at the corners. If they differ, the panels will not close.



The pipe pieces, back edge at the bottom. Only the top C gets the feed hole: 17.7 mm up from the back edge, halfway across. The strips above C and D are the front tabs.

Laying out one sheet

Everything for one horn fits on one 12 × 24 in sheet, with a 3 mm cutting gap between pieces and 4 mm clear of the sheet edge. Turn every second trapezoid round so the pair nest. Repeat on a second and third sheet for the other two horns.



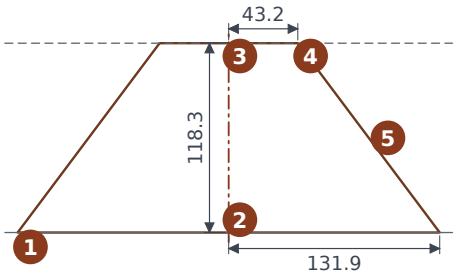
One horn's nine pieces on one sheet, to scale. They use 62% of it. Use the spare strip along the top to practise a fold and a soldered seam first.

How much copper to buy

One horn takes about 178 square inches of copper, tabs included. With room to cut between pieces, two 12 × 24 in sheets make two horns, not three: what is left over is too narrow for a third set of flare panels. Three horns need three sheets. A fourth is cheap insurance against a mis-cut.

Marking out a trapezoid accurately

Do not mark the slanted edges by measuring along them. Mark two parallel lines and a centre line, then measure out from the centre. Shown for A; B is the same with its own numbers.



1 Draw the long edge with a straight edge.
2 Mark its middle and square a centre line up from it.
3 Measure the height up the centre line and square a second line across there.
4 From the centre line, measure half of each edge out both ways along its own line.
5 Join the marks. Both slanted edges must come out 147.9 mm, on every A and every B.

Building it

1 Mark, cut and deburr

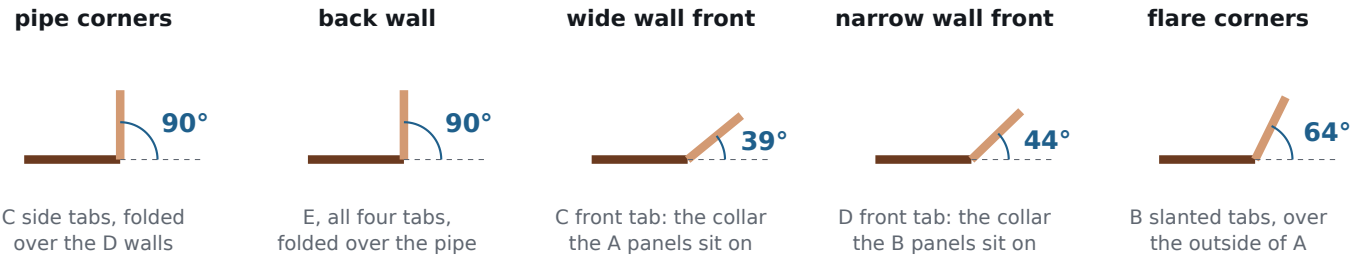
1. Mark every piece with a fine marker or a scribed line. Write its letter on what will be its outside face.
2. Cut with the snips, blade just outside the line. Cut the 45° ends of the B tabs and the corner notches of E last.
3. File every edge smooth. Burrs stop panels sitting flush, and they cut.
4. Flatten any piece the snips curled: lay it flat and press it with a block of wood. Check each piece against its size before going on.

2 Drill the feed hole while the top wall is flat

1. On the top C, mark a point 17.7 mm up from the back edge and 43.7 mm in from the side, which is halfway across. Centre-punch it.
2. Drill a small pilot, then open it to clear the connector's white insulator, usually 4 to 5 mm. Deburr both sides.
3. Hold the flange over the hole, centred and square to the edges, and mark its four holes through it. Drill them 2.7 mm.

3 Fold the tabs

Clamp the piece between two straight edges with the fold line exactly at their edge, then press the tab over with a block of wood.

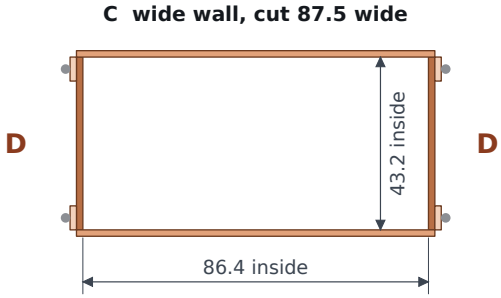


Cut a cardboard angle gauge for each of the three odd angles and hold it against every fold of that kind.

Which way each tab folds

C side tabs, E tabs and B tabs fold away from the outside face, so they wrap around the neighbouring wall. C and D front tabs fold toward the outside face, flaring out like the rim of a funnel: that rim is the collar the flare sits on.

4 Build the pipe



Looking into the back of the pipe

The narrow walls (D) stand between the wide walls (C). The wide walls lap over their edges, which is why they are cut 87.5 wide, not 86.4.

Each C side tab folds down over the outside of a D wall. Solder there. Grey dots: solder. Thickness drawn 3×.

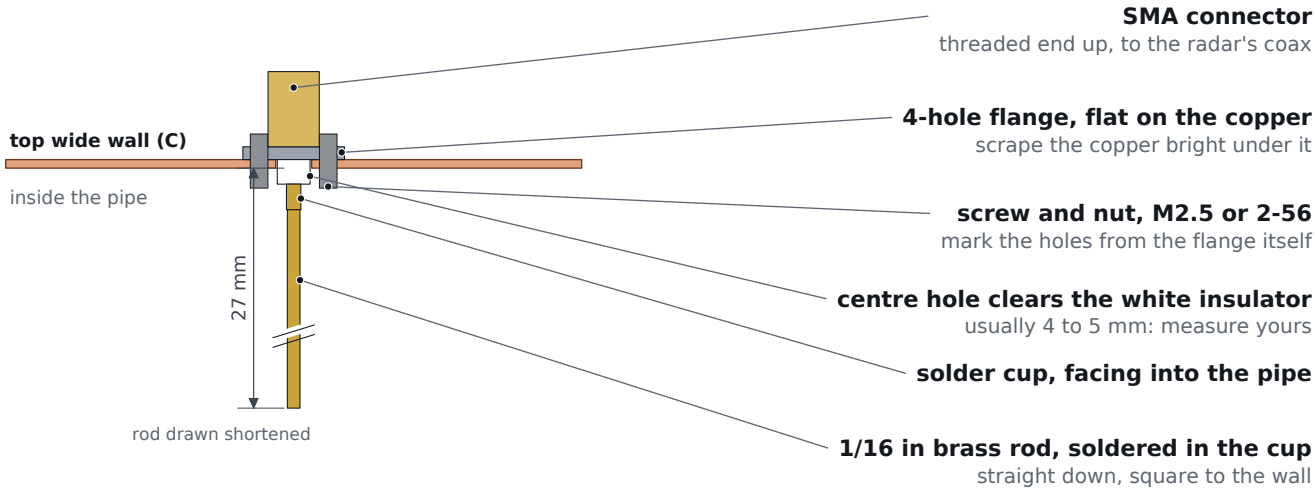
1. Stand the two narrow walls (D) on edge, 86.4 apart inside, and lay the bottom C across them, side tabs folded down over their outsides. Clamp.
2. Turn the box over and fit the top C, feed hole up, the same way.

3. Measure inside with calipers at both ends: 86.4 × 43.2, within half a millimetre, and the two diagonals equal. Adjust before soldering.
4. Tack each corner with a spot of solder, check again, then run solder along every tab from the outside. Keep the inside clean.

5 Close the back

1. Slide the back wall (E) over the back of the pipe, tabs forward over the outside. The pipe walls butt against its inside face.
2. Solder all four tabs. The inside face of E is what the probe is measured from, so it must sit flat and square across the whole end.

6 Fit the connector and the probe



Cut through the top wall at the feed hole.

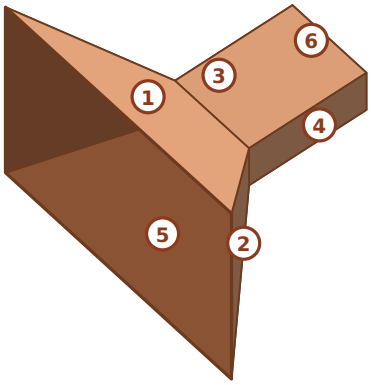
1. Cut 32 mm of brass rod and solder one end into the connector's cup, straight in line with it. If the rod will not enter the cup, file its end down until it does.
2. Scrape the copper bright around the hole. Push the rod in through the hole and bolt the flange down tight: it must touch metal all round.
3. Trim the rod so 27 mm stands inside, measured from the inside face of the wall. It must hang straight down, square to the wall.

> **The probe is the one part worth tuning**
>
> Its length and its distance from the back wall decide how much of the signal gets into the horn instead of bouncing back down the cable. The HFSS simulation sweeps both, and these numbers are its answer: the back wall closer than a quarter wavelength is deliberate. Within 1 mm either way the horn still matches. To trim the rod later, unbolt the connector and lift it out with the rod attached.

7 Add the flare

1. Set the collar against the cardboard gauges: the two C front tabs lean out 39° and the two D front tabs 44°.
2. Set the top and bottom panels (A) on the C collar tabs, short edge to the pipe. Tack each at the middle of its short edge only.
3. Set the side panels (B) on the D collar tabs, their slanted tabs over the outside of the A panels. Tack each at its short edge.
4. Pull the corners together and tack each corner seam at the mouth.
5. Measure the mouth at the rim: 263.8 × 193.1 inside, and the two diagonals equal. Squeeze or spread the corners until it is, then tack the middle of each corner seam.
6. Run the four corner seams and the collar, all from the outside, working from the pipe toward the mouth.

8 Check it



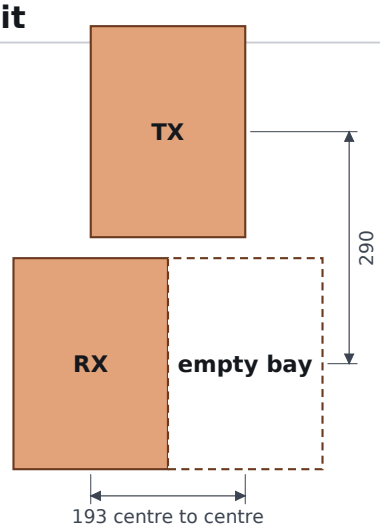
Every piece to every other piece
Meter on its lowest resistance range. Touch the probes to bare copper on two different pieces. Test 1 to 2, 1 to 3, 1 to 4, 1 to 5, 2 to 3, and on: every pair, across every seam.

Pass: under 0.5 Ω, every pair.

Then the connector (6): its outer body to the copper under 0.5 Ω, its centre pin to the copper open. The pin touching the copper is a short.

Then measure everything on the record page. Build all three horns before fitting any: the two receive horns should come out as close to twins as you can make them.

9 Mount it



Seen from the front, into the mouths

Turn every horn a quarter turn, so the 193.1 side runs across and the 263.8 side runs up. The connector ends up on the side of the pipe.

All three mouths flush in one plane. The RX horn and the empty bay touch, and TX is centred above the pair.

Grip each horn by its pipe, never by the flare.

Build record

Measure each horn and write the numbers in. This page is the evidence the horns were built to the design, and it is what the simulation results get compared with.

check	target	horn 1	horn 2	horn 3
mouth width, inside	263.8 ± 2			
mouth height, inside	193.1 ± 2			
mouth diagonals	equal ± 2			
flare depth, mouth to pipe	91.5 ± 1			
pipe inside, width × height	86.4 × 43.2 ± 0.5			
pipe length, back wall to flare	115 ± 2			
probe centre from back wall	17.7 ± 0.5			
probe length inside	27 ± 0.5			
probe square to the wall	check with a square			
every seam, piece to piece	under 0.5 Ω			
connector body to copper	under 0.5 Ω			
connector centre pin to copper	open			

Tuning, once the radar runs

No test instrument is needed. With the radar running, set it to a single frequency, put a corner reflector at a fixed distance straight in front, and watch how strong its echo is. Trim the receive horn's probe half a millimetre at a time and keep the length that gives the strongest echo. Then trim the spare horn until it reads within 1 dB of the one you fitted, so the pair match.

horn	length inside (mm)	echo strength	notes

What goes wrong

symptom	likely cause
the mouth will not close to size	the slanted edges of A and B differ. Measure both against 147.9 mm and trim the long one
a seam reads over 0.5 Ω	a dry joint, or flux trapped under a tab. Clean it and re-flow the solder
connector body open to the copper	oxide or flux under the flange. Scrape the copper bright and bolt it down again
connector centre pin shorted to the copper	the rod or cup touches the edge of the hole. Centre the connector over the hole
a weak echo that trimming will not fix	a gap somewhere in the metal. Repeat the continuity check on every seam